

MD. SHAHARIAR HASSAN RONOK

ECB Chattar, Manikdi, Dhaka 1206, Bangladesh | (+880) 1317-645665 | 346ronokarya@gmail.com
[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

RESEARCH & ACADEMIC INTERESTS

Wireless Systems & Security: Physical layer security, 6G wireless transport infrastructure, high-throughput packet processing (DPDK), and channel modeling.

Digital Systems & Microarchitecture: RISC-V microarchitecture, digital IC design & verification, FPGA hardware prototyping (Xilinx Artix-7), and pipelined datapaths.

Embedded Systems & Low Latency: Bare-metal C/C++ firmware, real-time operating systems (RTOS), and Linux kernel bypass (HugePages, core pinning).

EDUCATION

Rajshahi University of Engineering & Technology (RUET) — Rajshahi, Bangladesh

Bachelor of Science in Electrical & Computer Engineering (ECE) | Jan 2018 – Oct 2023

Cumulative GPA: 3.21 / 4.00 | Dual Track Specialization in Electrical & Electronic Engineering (EEE) and Computer Science & Engineering (CSE)

Undergraduate Thesis: "Analyzing Secrecy Performance in Wireless Communication Networks with Randomly Distributed Eavesdroppers"

Supervisor: Milton Kumar Kundu, Assistant Professor, Department of ECE, RUET. Evaluated secrecy outage probability and achievable secrecy rates under spatial Poisson point distributions using stochastic geometry and wireless fading channel models.

Undergraduate Review Paper: "A Review on Wearable Technologies in Early Detection of Lymphedema"

Department of Electrical & Computer Engineering, RUET. Surveyed non-invasive wearable biosensing architectures (force-sensitive resistor cuffs, epidermal dielectric/strain LC patches, and ultrasonic transducers) for continuous monitoring, and evaluated growth models and change-point algorithms for early detection.

Core Coursework: VLSI Design, Computer Architecture, Digital Logic Design, Microprocessors & Microcontrollers, Embedded Systems Design, Analog Electronics.

STANDARDIZED TESTS & LANGUAGE PROFICIENCY

IELTS Academic: Overall Band 6.5 (Listening: 6.0, Reading: 6.0, Writing: 6.5, Speaking: 7.0) — Test Date: September 2026

TECHNICAL SKILLSET

High-Speed Networking & Systems: DPDK (Data Plane Development Kit), Kernel Bypass, HugePages Allocation, CPU Pinning, Core Isolation, POSIX Shared Memory (SHM), TAP Interfaces, dpdk-testpmd.

Digital Design & Architectures: Verilog, SystemVerilog fundamentals, 3-Stage RISC-V Microarchitecture, Hazard Detection & Forwarding Logic, Synchronous FIFOs, Moore & Mealy FSM Controllers, ALU Design.

Verification & Automation: Cocotb (Python Testbenches), Verilator, Icarus Verilog, GTKWave, GDBWave, Microarchitectural Assertions, Makefile Test Automation, GitHub Actions, GitLab CI, Docker.

EDA & FPGA Prototyping: Xilinx Vivado (Behavioral Simulation, Synthesis, Implementation, Timing Constraints/XDC, Hardware Manager), Artix-7 FPGA Implementation.

Embedded & Firmware: Bare-Metal C, STM32 (ARM Cortex-M), AVR/Arduino, ESP32, UART, SPI, I2C, gRPC (Python/C).

PCB & Lab Equipment: KiCAD, Multisim, Proteus, Digital Storage Oscilloscopes (DSO), Logic Analyzers.

PROFESSIONAL ENGINEERING EXPERIENCE

Siliconova — Dhaka, Bangladesh

Embedded Software Engineer I | July 2024 – Present

1. Develop high-throughput data plane software in C using DPDK to process network packets at line rate with low latency.
2. Configure Linux servers using HugePages, CPU core pinning, and POSIX shared memory to enable zero-copy packet processing pipelines.
3. Write and test Verilog RTL modules on Xilinx Artix-7 FPGAs to stream packetized data over Ethernet to host servers.
4. Work with FPGA implementation flows in Vivado, covering RTL simulation, timing closure using XDC constraints, and programming bitstreams onto physical boards.
5. Build gRPC services in Python and C to manage hardware registers, update link settings, and monitor system telemetry.
6. Containerize test environments using Docker and automate network testing with TAP interfaces and dpdk-testpmd inside CI/CD pipelines.

SELECTED HARDWARE & RESEARCH PROJECTS

3-Stage Pipelined RISC-V Core & Automated Cocotb Regression Suite

Siliconova VLSI Track | **Tools & Stack:** Verilog, Cocotb, Verilator, GTKWave

1. Converted a single-cycle PicoRV32 processor into a 3-stage pipelined core (IF, ID/EX, MEM/WB) using Verilog.
2. Implemented a Hazard Detection Unit to handle 1-cycle load-use dependencies along with combinational data forwarding logic.
3. Integrated branch-flushing logic within the decode stage to clear speculatively fetched instructions on branch mispredictions.
4. Wrote cycle-accurate Python testbenches in Cocotb for directed instruction tests and set up automated regression runs using Makefiles.

Digital Logic Blocks & Verification Suite

Academic & Open-Source Track | **Tools & Stack:** Verilog, Cocotb, Icarus Verilog

1. Designed a configurable synchronous FIFO in Verilog with full, empty, almost-full, and almost-empty flag indicators.
2. Tested simultaneous read/write operations using Cocotb to verify pointer synchronization, overflow/underflow logic, and data accuracy.

3. Designed Moore and Mealy FSM controllers with glitch-free registered outputs for serial bus communication protocols.

Microcontroller System Board & Multi-Sensor Interface

RUET Academic Capstone | **Tools & Stack:** KiCAD, Multisim, Bare-Metal C

1. Designed schematics and a 2-layer PCB layout in KiCAD with split ground planes for low-noise sensor performance.
2. Verified power decoupling, clock integrity, and signal noise levels using digital oscilloscopes in the lab.
3. Wrote non-blocking, interrupt-driven bare-metal C drivers for UART and SPI to stream multi-sensor data.

ACADEMIC REFEREES

- **Milton Kumar Kundu**

Assistant Professor, Department of Electrical and Computer Engineering (ECE), Rajshahi University of Engineering & Technology (RUET)

Email: milton.kundu@ece.ruet.ac.bd | Role: Undergraduate Thesis Supervisor

Research Focus: Wireless Communications, Physical Layer Security & Network Modeling

- **Md. Robiul Islam**

Assistant Professor, Department of Electrical and Computer Engineering (ECE), Rajshahi University of Engineering & Technology (RUET)

Email: robiul@ece.ruet.ac.bd | Role: Academic Instructor / Department Referee

Research Focus: Embedded Systems, Digital Systems & Signal Processing